

Sustainable Design Engineering

HMW
Rethink how we design aircraft seating to tight safety regulations, in a way that hands-on stakeholders directly benefit from sustainable solutions?

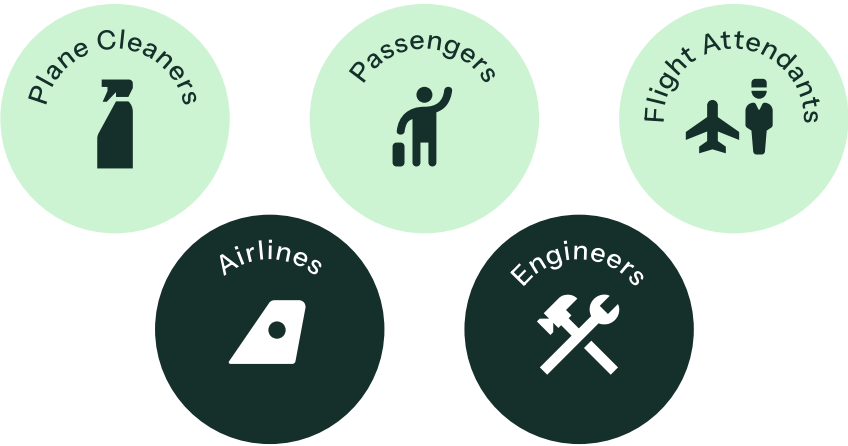
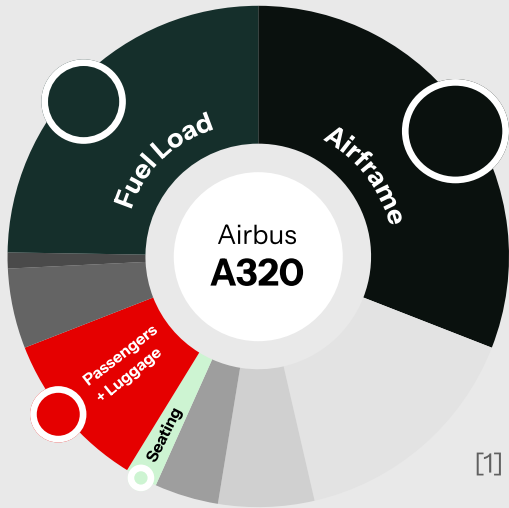


Project Context

Research Summary

Weight

Seating only accounts for **2% of an aircraft's takeoff weight**. Other more prevalent weight factors mean that seats are **not a priority** to an airline, or engineer. Airlines aim to **maximise passenger count**, leading to an increase in seating capacity per plane [2], which drives seats to be thinner and lighter.



Stakeholders

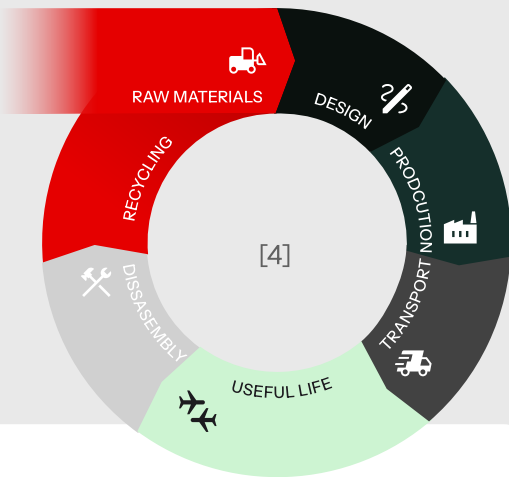
There were 5 main stakeholders identified as effected by seating, with a clear divide between those with **high and low 'contact hours'**. The most prominent 'hands-on' stakeholders were **plane cleaners, passengers and flight attendants**.

Despite the proportionally high time spend with seating, we found that **cleaners and flight attendants** were often **overlooked** [3].

Flow Analysis

Analysing product flow for the system highlighted several pain points. The most dominant being the design phase, where **time and financial pressures** from airlines push manufacturers to **optimise for cost**. In turn **encouraging planned obsolescence** as seat manufacturers need to repay r&d costs [3].

Disassembly also presented significant issues, the teardown of the Weber design demonstrated the impact of **high component count** and **lack of DfD**. Comparison to the Doy Design 'Ultra Slim' showed **benefits in minimisation** techniques but it retained the same slow-to-disassemble frame-seat-cover construction system [5].



Initial Problem Statement

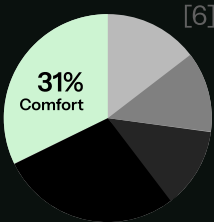
Although seating is **only 2%** to an airline or engineer, it makes up a **clear majority** of a **passenger, cleaner or attendant's time**. These key stakeholders often don't 'see' the direct benefit of sustainability.

Rigorous regulations have **restricted innovation**, leading to stagnation and single-track optimisation of **only weight**. Can sustainability benefit from **re-evaluation of regulations** and challenging traditional component make-up.

Further Research

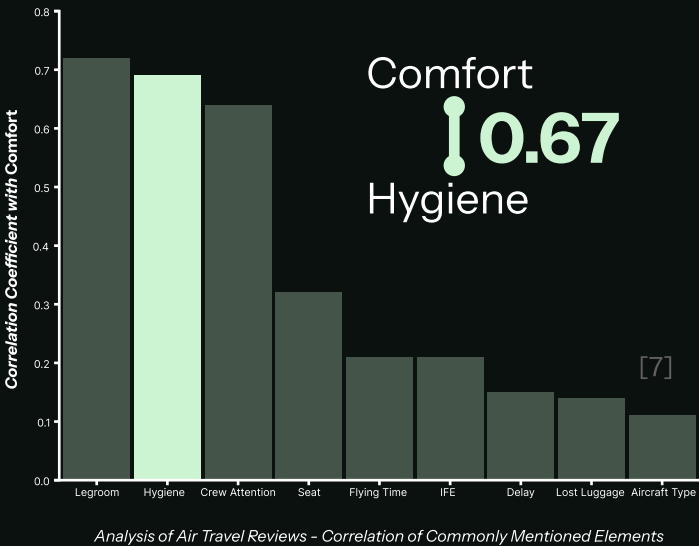
Passenger Comfort

Group user research showed comfort had the **greatest influence** on a **passenger's experience** over factors like accessibility and functionality.



Hygiene is one of the most important factors when it comes to passenger comfort [7]. We need to place **significance on cleanliness** as well as more typical physical and ergonomic interpretations of comfort.

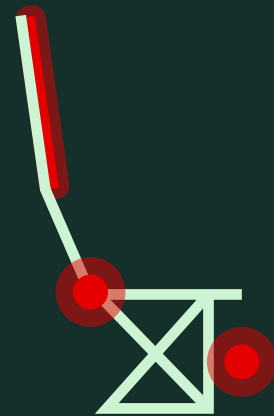
Other top factors include Legroom (*ergonomic*) and Crew Attention, with **significant margin** over elements like the In-Flight-Entertainment and the Seat itself.



Ergonomics

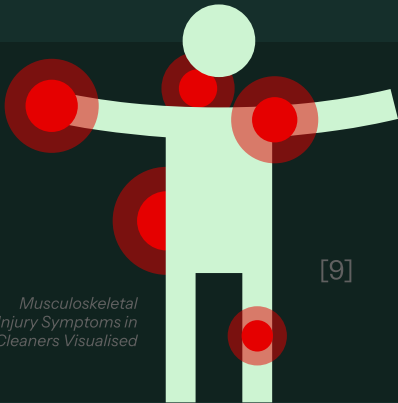
Passengers

In the survey that was conducted in the initial report, we found that passengers' most important seat features were **Seat Material, Legroom and Recline Angle** [6]. Over the last 40yrs, fuelled by the desire to increase capacity, airlines have **decreased seat width 5cm, legroom by 11cm** and have implemented 'fixed recline' configurations [8]. This simply **restricts and removes** elements of comfort that passengers desire.



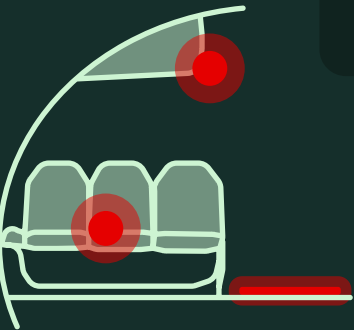
Plane Cleaners

Ergonomics for plane cleaners are almost completely overlooked, although cleaning involves *repetitive movements, high force requirements and extreme, static or constrained postures* [9]. Over 1 year, **83% of cleaners** present symptoms of **musculoskeletal injury**; most commonly in the *lower back, hand/wrist and shoulders*.



Flight Attendants

Similarly to plane cleaners, flight attendants suffer heavily from to ergonomic issues. Musculoskeletal injury is common, often stemming from **reaching for overhead lockers, thin-isles with heavy service trolleys, and leaning over to access passengers** [10]. In addition, flight attendants also face other increased injury risks, including lack of restraint in turbulence, which makes injury **12X more likely** than for passengers [11].



HMW

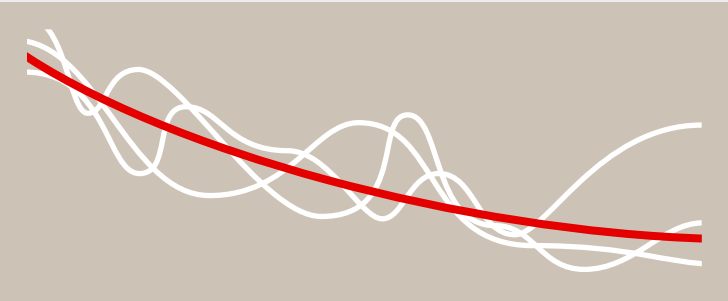
How might we challenge the conventional construction of airline seating systems to **maximise user comfort**. Benefiting the **hygiene and ergonomics of passengers, flight attendants and plane cleaners**?

Key Stakeholders



‘Hands On’ Stakeholders

With a focus on comfort, it the ‘non-contact’ stakeholders become less relevant. Therefore, Plane Cleaners, Passengers and Flight Attendants took primary focus.



Time vs Comfort

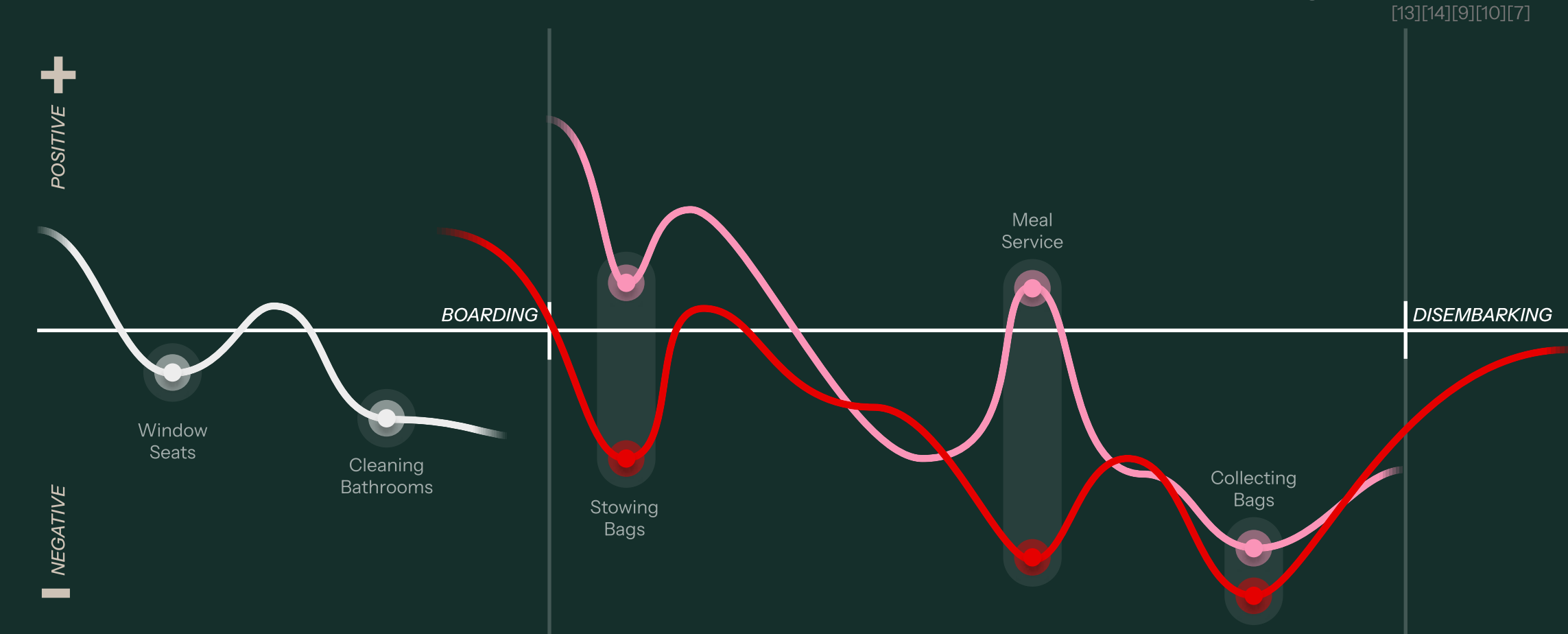
Hygiene can be a major part of why the general trend of comfort with time is down, as people tend to notice/take in their surroundings over a long period, becoming aware of poorly kept environments [12].
This graph is constructed from the comfort tracking graph.



Tackling Hygiene

The London Underground employs a few tactics to tackle hygiene. The biggest is the loud moquette patterns used to hide dirt. They also use extremely simple surface geometries for fast and effective cleaning.

Tracking Comfort



Plane Cleaners

INFO

AGE

38

PAIN POINTS

- Speed, cabin cleaners operate under strict time constraints so need to move quickly and be efficient
- Task Repetition, a high number of seats and identical movements as they move through the cabin
- Moving in and out of seats. Especially window seats as they have to sit and shuffle to reach all of the surfaces near the window area
- The most time consuming part is wiping the back of the seat with the tray table due to the complex surface geometry.
- ‘Hidden areas’ where items can be left behind from passengers
- Bathrooms are a large time sink, they are complex and require a higher standard of cleanliness

Passengers

INFO

AGE

30

PAIN POINTS

- Bag Storage, often cumbersome and slow, makes boarding a big time sink.
- This is also true for disembarking, and can be amplified by the desire to leave the aircraft upon landing.
- General lack of space, passengers specifically have problems with headrests and legroom.
- Lack of options, people bring a lot of items on flights, and need places to put them without sacrificing their personal space.
- Comfort over time, they want to be able to relax/ rest and pass the time rather than be actively engaged.
- Standard of Cleanliness, passengers want the plane to satisfy an overall level of hygiene.

Flight Attendants

INFO

AGE

43

PAIN POINTS

- Flight Attendants views are often obstructed, they need to be able to communicate and identify passengers
- Movement speed is also valued, they need to be able to traverse the cabin quickly in case of emergency. They often lack grip points when moving, which is especially important in turbulent conditions
- Moving in and out of seats, attendants have to be able to access every person on the flight and repeated bending causes spinal problems.
- Pushing the trolley, they are heavy, awkward to move and often obstructed by passengers.
- There is also a time pressure on attendants, they need to ensure the passengers are prepared for a strict schedule.

Design Requirements



DESIRES

- Fast, unobstructed cleaning with minimal components to handle.
- Surfaces and materials that are smooth, hygienic, and easy to visually assess for cleanliness.
- Seat structures without hidden or hard-to-reach areas where debris can accumulate.
- Ergonomic access to all seat areas to reduce repetitive strain and minimise awkward movements.
- Efficient waste removal processes and clear pathways that support rapid turnaround times.



DESIRES

- A reliably clean, hygienic environment on first impression and throughout the flight.
- Adaptable personal space that supports long-term comfort (legroom, headrest, posture).
- Intuitive storage that speeds boarding, reduces clutter, and keeps personal items accessible.
- Aesthetically modern seating that feels premium and well-maintained.
- Smooth boarding and disembarking that supports fast movement through the cabin.



DESIRES

- Unrestricted movement throughout the cabin, when operating trolleys or assisting passengers.
- Ergonomic access to seated passengers without excessive bending or reaching.
- Clear visibility across the cabin and seating layouts that support quick passenger interaction.
- Stable, integrated gripping or support points along the aisle for safe movement.
- Seat designs and cabin layouts that reduce delays and support time-critical service tasks.

REGULATION REQUIREMENTS

Drawn from group and initial broad individual research.

REGULATIONS - EASA

- Flammability
- Toxicity
- Structure
- Energy Absorption
- 'Material Suitability'

Specification Points

Overall Project Goal: OPTIMISING FOR COMFORT

By optimising for comfort I aim to find solutions that that benefit a broader cohort of hands-on users, focussing on the social sustainability of those who interact directly with seats.

1

.1

Seat structures must eliminate concealed or hard-to-access areas, allowing all surfaces to be cleaned without entering awkward positions.

.2

Cabin layout and seat geometry must allow cleaners to move between rows quickly, without needing to sit or kneel inside passenger spaces.

.3

Per-seat clean time ≤ 90 seconds (includes tray, back of seat, and armrest).

.4

All passenger-facing surfaces are wipe-clean in a single cloth pass. No gaps or cavities > 6 mm where debris can lodge.

2

.1

Integrated personal storage must be accessible from a seated position and hold all typical carry-on essentials without obstructing leg or knee space.

.2

Materials and finishes must convey visual cleanliness and modernity, resisting wear that would make the seat appear dated or unhygienic.

.3

Seat layout must support efficient passenger flow, ensuring boarding and disembarking paths remain unobstructed and quick to navigate.

.4

Sustained comfort: no more than 10% increase in perceived discomfort after 3 hours (subjective scale).

3

.1

Integrated, unobtrusive grip points must be included along aisle-side seating, enabling stable movement during turbulence and service.

.2

Seat layout must preserve sight-lines across the cabin, allowing attendants to visually monitor passengers from standing positions at typical aisle points.

.3

Aisle clearance permits standard service trolley passage with ≥ 100 mm total clearance.

.4

Seat design enables passenger-service reach within ergonomic limits: ≤ 350 mm lateral reach from aisle.

4

.1

Structure of the seat must pass or clearly demonstrate the potential and intention to pass EASA 'Structure' and 'Energy Absorption' crash testing regulations.

.2

The materials used in the seat must pass or clearly demonstrate the potential and intention to pass EASA 'Toxicity', 'Flammability' and 'Material Suitability' regulations.

Initial Ideation

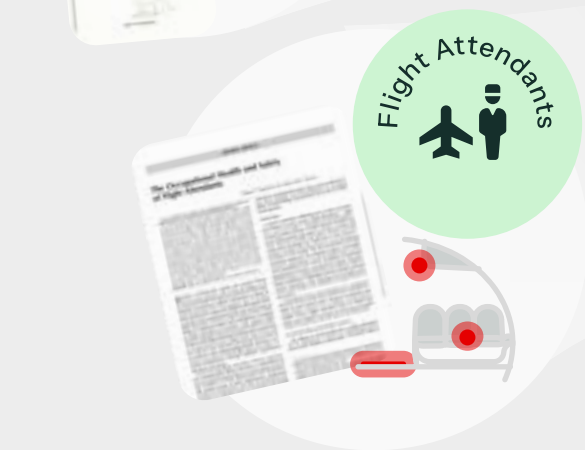
Passengers



Tony Seville



Flight Attendants

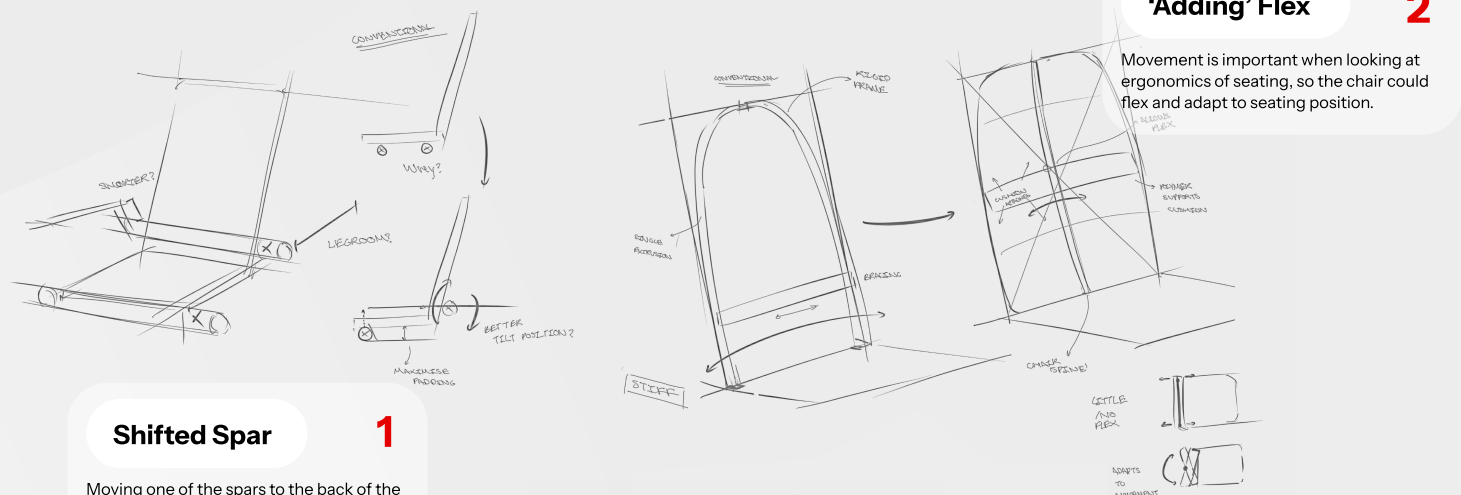


Plane Cleaners



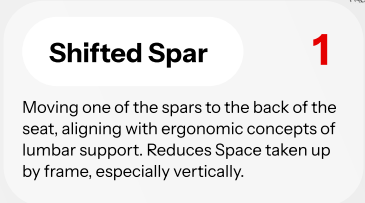
'Adding' Flex 2

Movement is important when looking at ergonomics of seating, so the chair could flex and adapt to seating position.



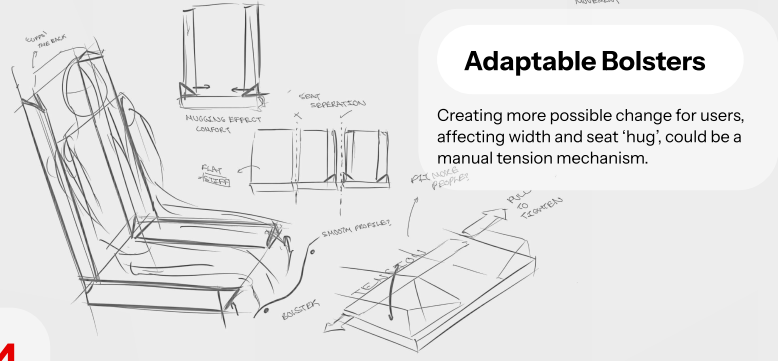
Shifted Spar 1

Moving one of the spars to the back of the seat, aligning with ergonomic concepts of lumbar support. Reduces Space taken up by frame, especially vertically.



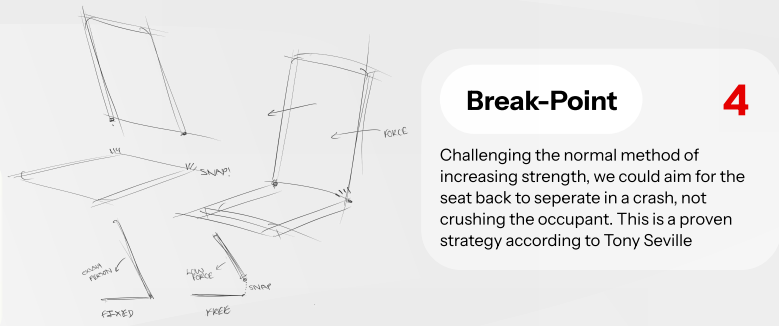
Adaptable Bolsters 3

Creating more possible change for users, affecting width and seat 'hug', could be a manual tension mechanism.



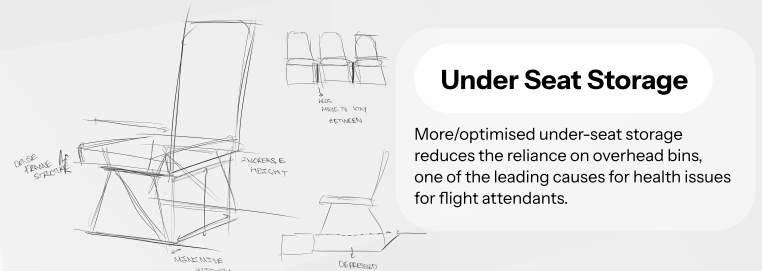
Break-Point 4

Challenging the normal method of increasing strength, we could aim for the seat back to separate in a crash, not crushing the occupant. This is a proven strategy according to Tony Seville



Under Seat Storage 5

More/optimised under-seat storage reduces the reliance on overhead bins, one of the leading causes for health issues for flight attendants.



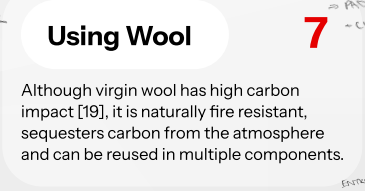
Seat Fabrics 6

Reduce the need for foam/padding, attaching straight to frame. This also allows 90deg bend without multiple parts, great for cleaning and spine ergonomics.



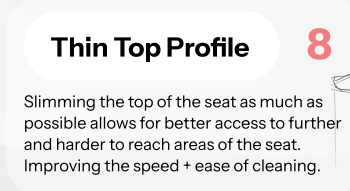
Using Wool 7

Although virgin wool has high carbon impact [19], it is naturally fire resistant, sequesters carbon from the atmosphere and can be reused in multiple components.



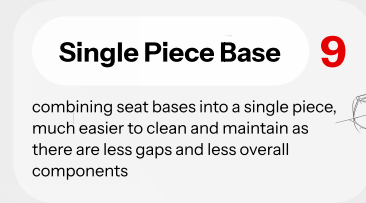
Thin Top Profile 8

Slimming the top of the seat as much as possible allows for better access to further and harder to reach areas of the seat. Improving the speed + ease of cleaning.

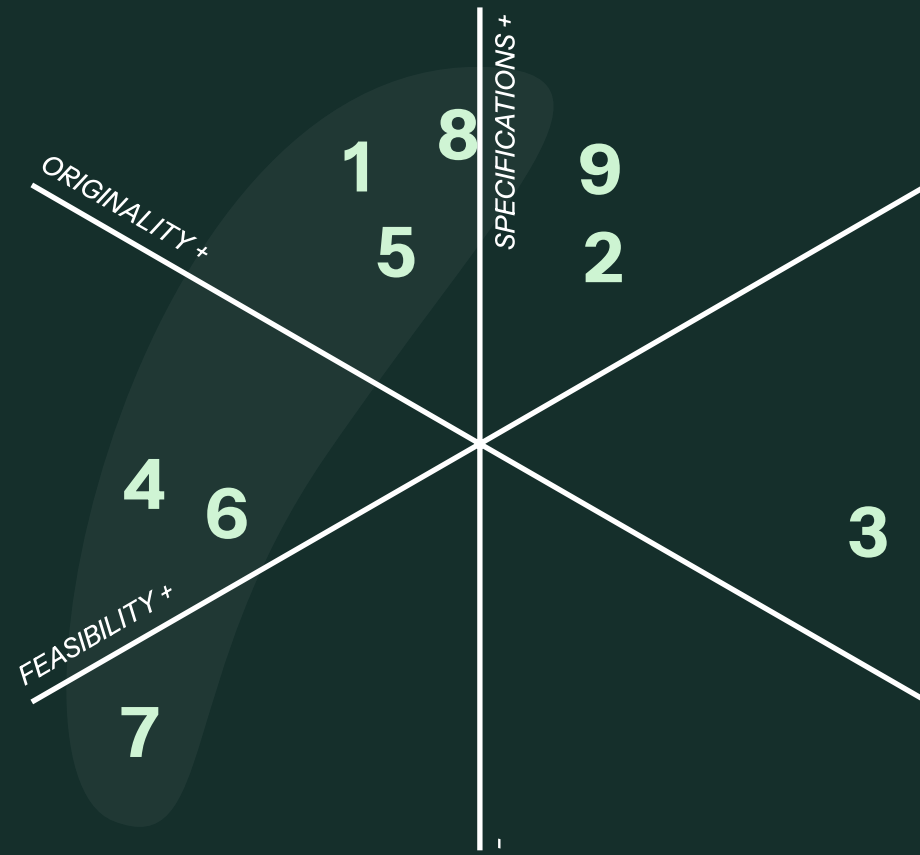


Single Piece Base 9

combining seat bases into a single piece, much easier to clean and maintain as there are less gaps and less overall components



Evaluation



Mapping

By plotting my concepts, in terms of feasibility, originality and 'closeness' to my specifications, I was able to identify which concepts stood out.

Although I did not use numeric analysis, comparing each concept to my design specification allowed me to see how combining concepts could be used to benefit multiple key stakeholders.

1,4,5,6,7,8

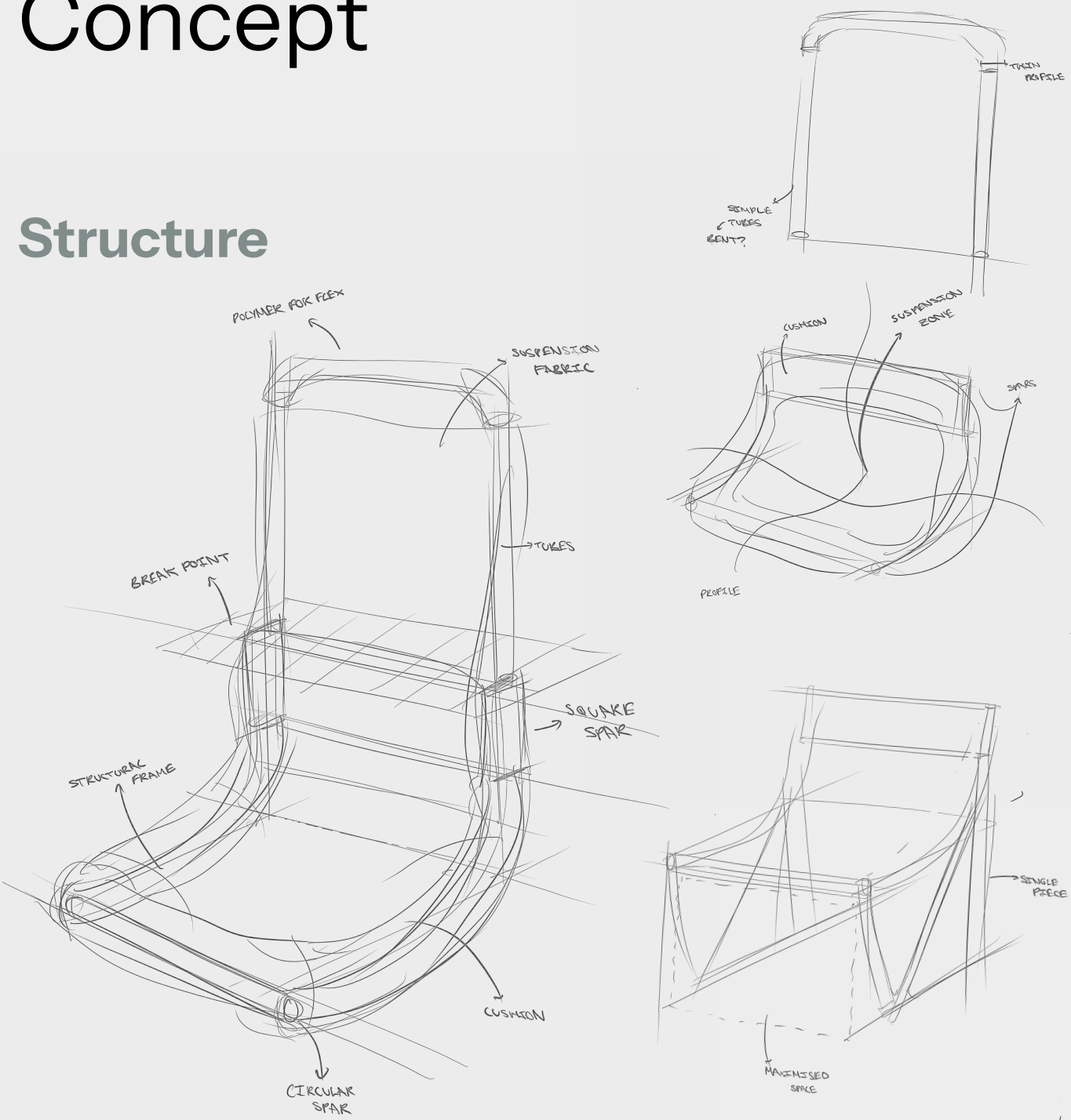
Moving Forward

Wanting to create a seat that benefits the comfort of all of my key users, I opted to combine several of these concepts.

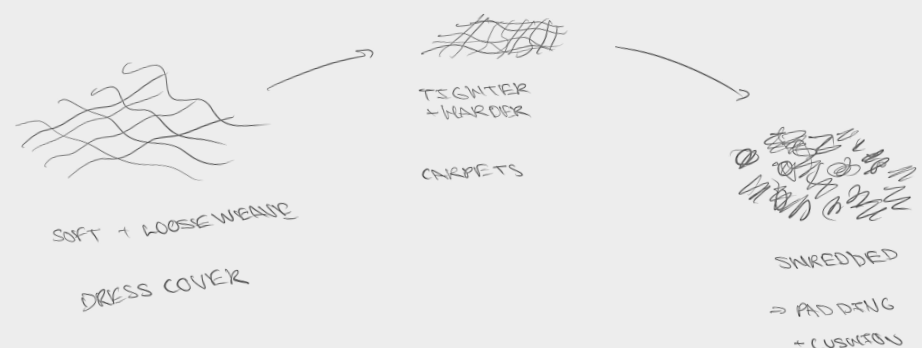
Many of the ideas work well in unison, and I decided to take forward many of the ideas that could benefit each other. For example, concepts 5 and 6 (*Under Seat Storage* and *Suspension Fabrics*) are mutually beneficial, with a reduction in foam 'freeing up' more space underneath the seat.

Selected Concept

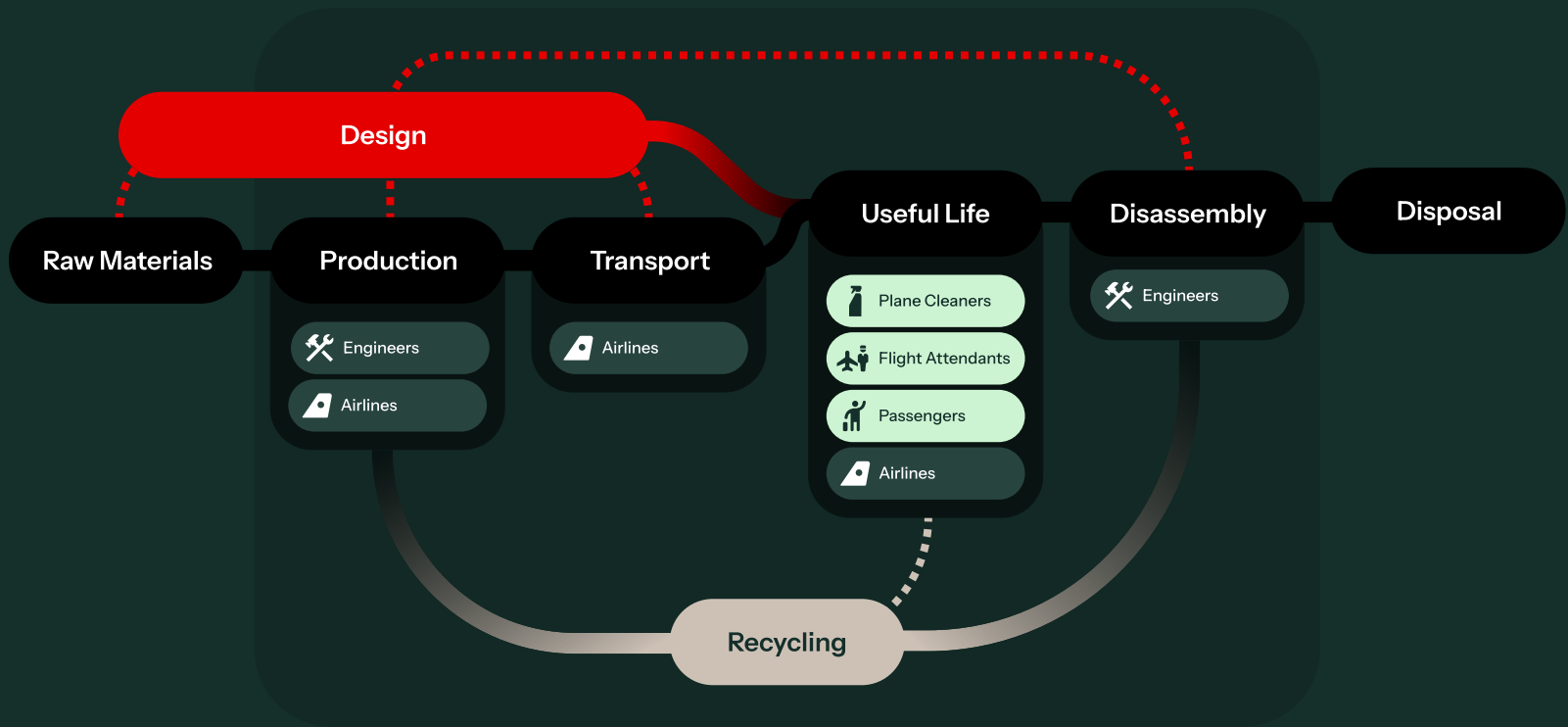
Structure



Fabric



System Level Change



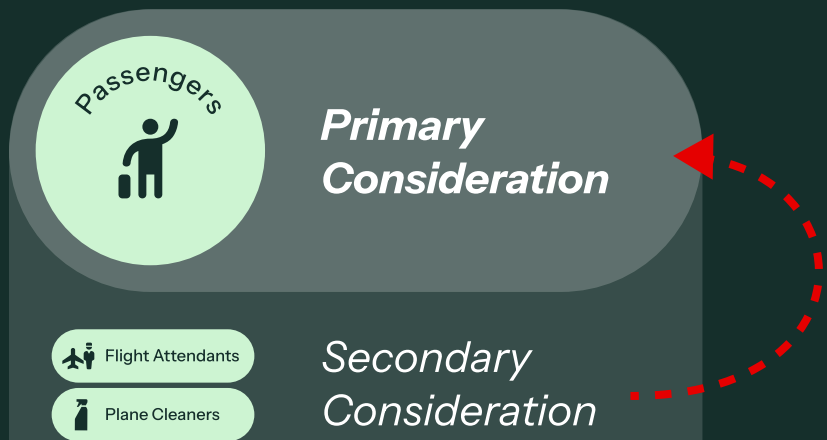
Focussing on Use Phase

The impact of the use phase is extreme, it is by far the highest ranking period for both emissions and effect on direct contact stakeholders.

Although this solution does not create significant shift in the life cycle of aviation seating, it aims to find solutions that more broadly benefit these 'hands-on' users.

[18]
99.8%
CO₂ + Embodied Energy

Changing the Design Process



Design User Priority

My concept aims to bring typically secondarily considered users into the forefront, 'shifting' them up the priority list.

By improving and optimising for these 'non-primary' users, we create solutions that benefit all users, including passengers.

Concept Development

HMW Reduce Window Seat Clean-Time

Plane Cleaners

Extending the cushion to the edge of the airframe structure helps to create one unified surface, improving ease and speed of cleaning.

This is also aided by minimisation of gaps between seats, reducing entrapped dirt and simplifying the cleaning process

HMW Use Wool as Suspension Fabric

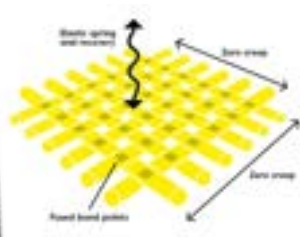
Wool cannot function as a suspension fabric alone, so needs to be combined with an elastomer. Threading elastic fibres into a woolen weave is lighter but makes recycling impossible, so using 2 separate layers is preferred.

Both Herman Miller and Arville use DuPont Hytrel (TPEE) filaments for elasticity and durability.

56% Elastomer



"Elastomeric Monofilaments"

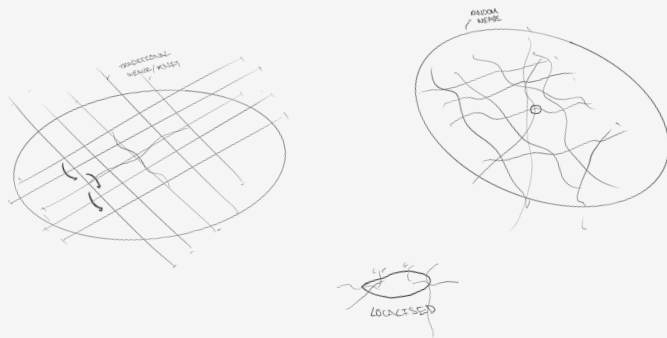


100% Polyester

HMW Increase Dress Cover Durability



Extremely light and thin, uses a pattern of stronger threads to stop tearing



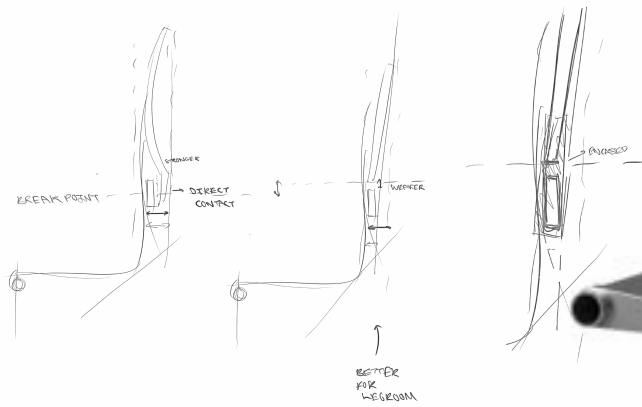
Using a random knit pattern can stop broken threads from spreading, 'localising' the damage.

Threading in stronger fibres is great for durability but a nightmare for recycling, especially combining natural and synthetic.

HMW Reduce the Impact on Legroom

The transfer (break-point) between the upper and lower structures has a large impact on the seat behind.

To improve legroom, we can use a smaller contact area, mounting the tubes to the top of the frame. This is a weaker connection but can easily be reinforced at the joint.



Passengers

HMW Create a 'Nudge' Towards Clean Aisles



Flight attendants have problems both with overhead stowage and moving up and down the aisle. To combat this, we can try to encourage passengers to keep the aisles clear:

Modifying the railing to terminate into the ground creates more of a guard between the aisle and under-chair space.

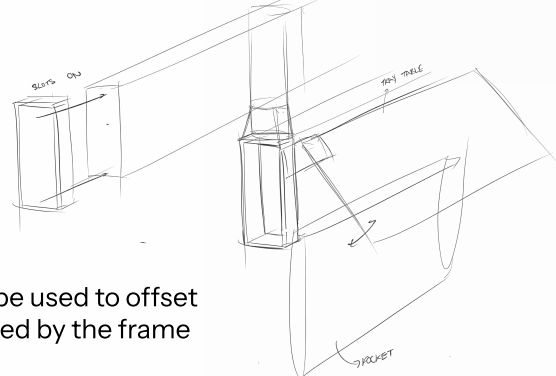
Using gentle lighting and segmenting the individual seats as much as possible, we can create a feeling of personal space and comfort [ref]. Therefore 'nudging' passengers to keep their belongings in the areas that feel most safe and secure.

Flight Attendants

HMW Minimise Component Count

By using a single moulded polymer piece to hold the upper frame to the lower, it's possible to adapt this piece to hold accessories + other essential components without adding to the component count.

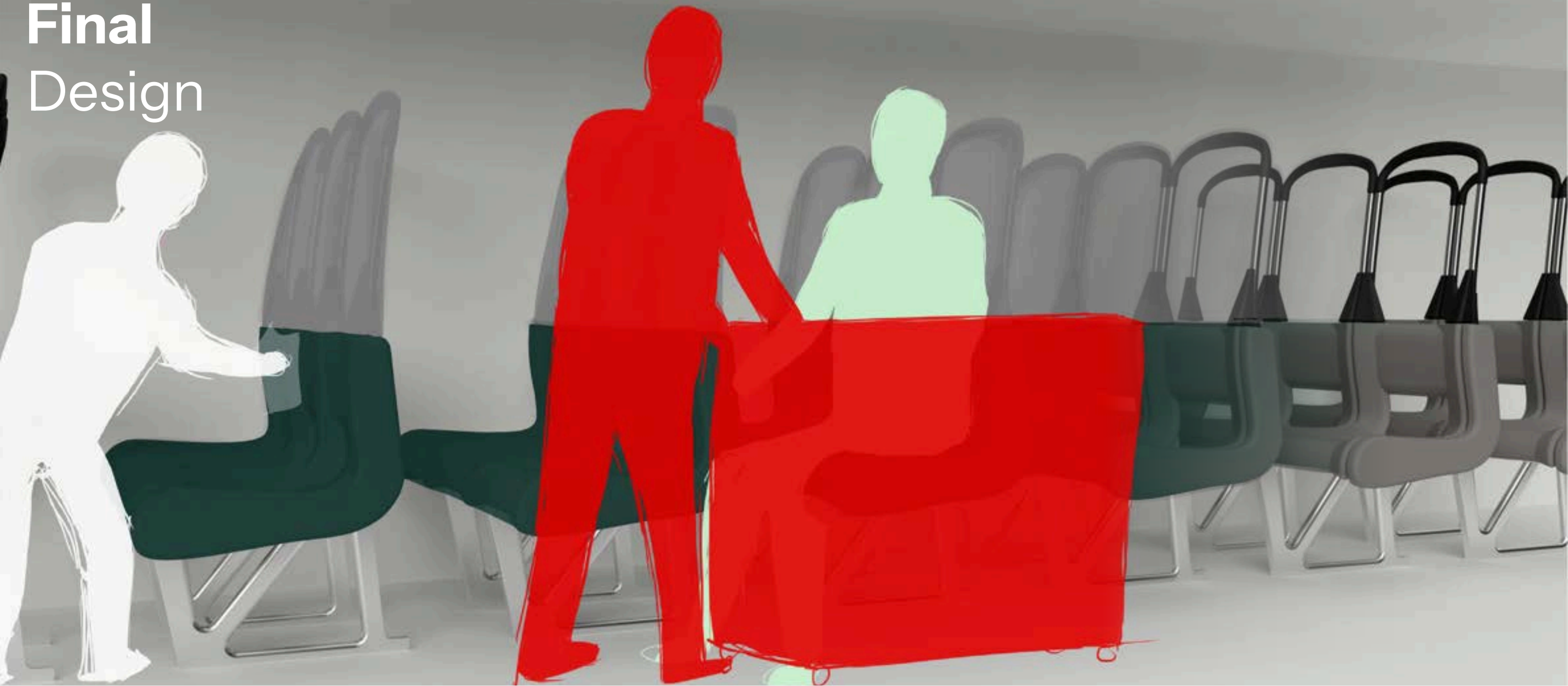
This greatly aids disassembly and be used to offset the additional complexity introduced by the frame changes.



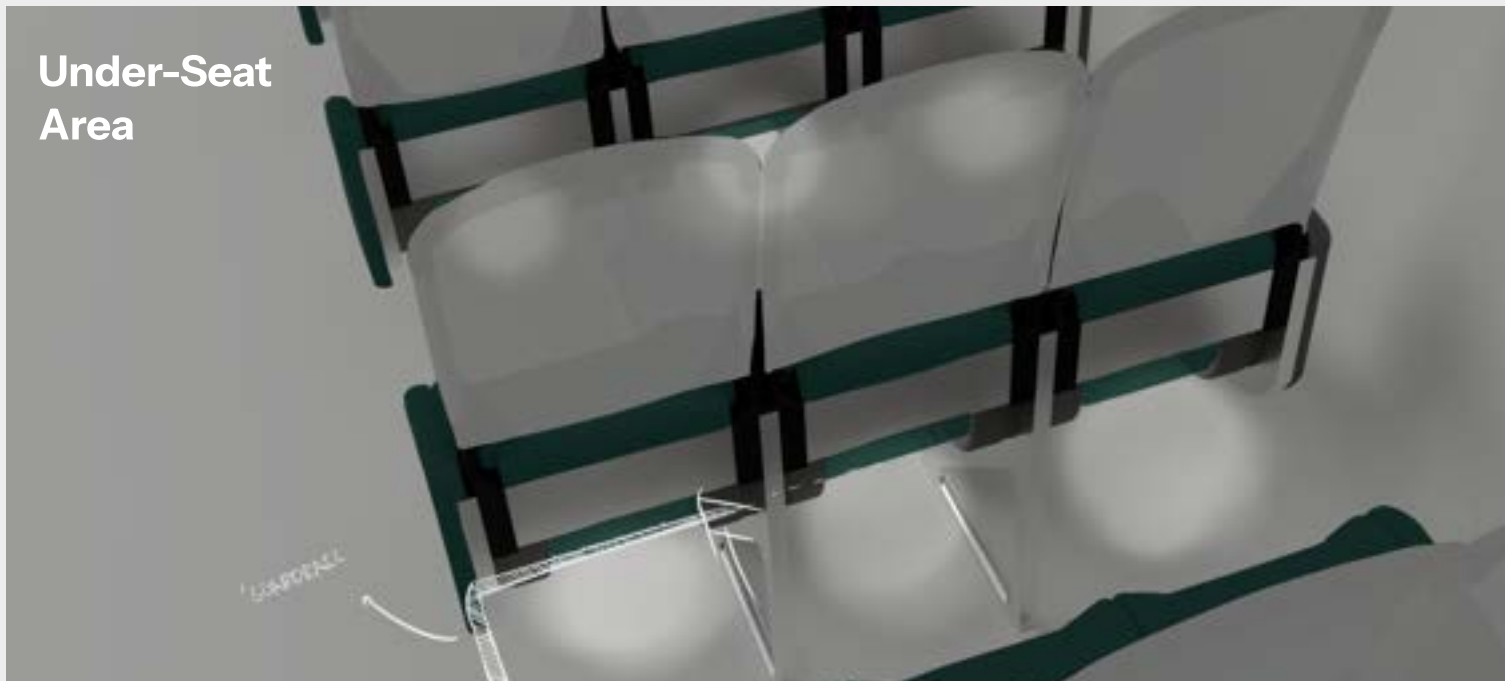
Gary Doy

7

Final Design



Under-Seat
Area



Structural
Components

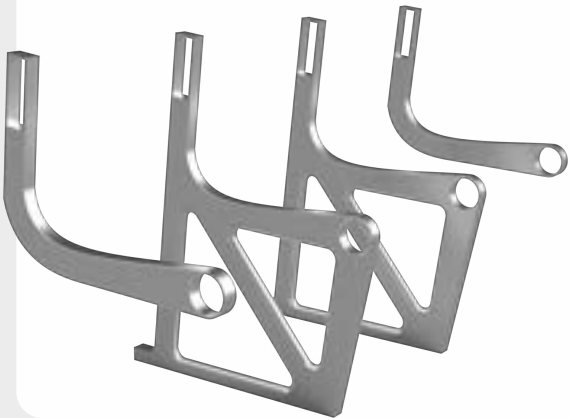


Design Breakdown

- Flight Attendants
- Passengers
- Plane Cleaners



Lower Frame



Material:
Aluminium
7050* - optimised for additive manufacture

Manufacturing Process:
Additive
Powder-bed Fusion at 'infill' density of 50%

User Benefit:

Passengers

Plane Cleaners

Thin profile with wide beams for better under seat zoning, as well as a completely smooth surface for easy cleaning.

Cushion



Material:
Recycled Wool
Shredded, and enclosed in further wool fabric for structure

Manufacturing Process:
Shredded, compressed and packaged
Further fabric - standard knit process

User Benefit:

Passengers

Thin profile with wide beams for better under seat zoning, as well as a completely smooth surface for easy cleaning.

Upper Frame



Material:
Aluminium
6000 series

Manufacturing Process:
Cutting to length
Single Pipe bend for shape

User Benefit:

Flight Attendants

Extreme thin profile for easier access and visibility of passengers.

Dress Cover



Material:
Recycled Wool
Elastomer (TPEE)

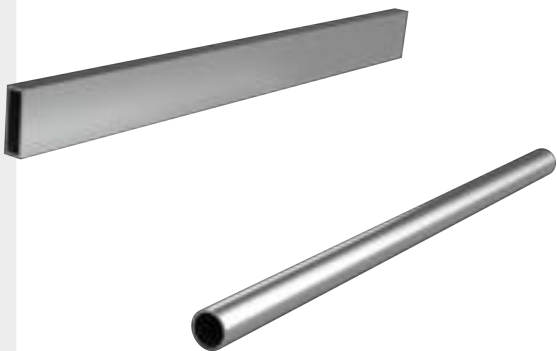
Manufacturing Process:
Wool - Random Weaving Technique
Weaved elastomer with fused strands

User Benefit:

Plane Cleaners

Smooth single surface with minimal gaps for fast and effective cleaning.

Spar System



Material:
Aluminium
6000 series

Manufacturing Process:
Standard Extrusions cut to length

User Benefit:

Passengers

Change in arrangement brings improved spinal ergonomics and helps to optimised the under-seat space.

Polymer Fixings



Material:
Polypropaline
6000 series

Manufacturing Process:
Injection Moulding

User Benefit:

Passengers

Plane Cleaners

Minimal/combined components make for simple cleaning tasks. Use of polymer in joining areas allows for ergo flex.

Concept Evaluation

Against Specification

Point	Hit?	Point Relevance?
1.1	HIT	YES
1.2	NOT	YES
1.3	UNSURE	YES
1.4	HIT	YES
2.1	HIT	YES
2.2	HIT	--
2.3	UNSURE	--
2.4	NO	YES
3.1	UNSURE	--
3.2	HIT	YES
3.3	NO	YES
3.4	UNSURE	YES
4.1	HIT	YES
4.2	HIT	YES

Summary:
Overall, although some specification points were not met, it is clear that this concept made significant progress towards satisfying the desires of my key users, especially plane cleaners and flight attendants

Eco Audit

Seat Weight



Though this concept is not competitive with the Doy Ultra Slim, it is in line with a modern economy seat, clearly beating out the Weber design that we were able to teardown.

Non-Use Phases

Despite 'Use' being the clearly dominant phase, it is still important to consider the production, transport and end of life of the product.

PRODUCTION

The largest difference is in this section, with the wool components of my concept alone being **51.385kgCO₂E***

*if assumed virgin material

TRANSPORT

Tracking with the weight figures, the transport phase for my concept aligned closely with the benchmark seats.

END OF LIFE

Use of down-cycled and recycled fabrics **offset** the larger environmental costs of production.

The Impact

Still **2%**



Relatively little impact.....
With current airline seating roughly matching my concept in terms of weight, airline seats remain around 2% of the total takeoff weight.

Summary:
My concept does not significantly reduce the environmental impact of airline seating in terms.

However, it is important to consider the social sustainability impacts.

Concept Validation

Business Feasibility

Gary Doy

The current industry business model is selling the seats at a loss, with accessories and components that are designed to break/be replaced. Those components are then sold at high prices to make back the money that is lost on the primary seating components.

Daniel Clucas

Another problem in design is that the other stakeholders aren't willing to commit to a **long term seating solution** either, as seen with the loop concept, airlines specifically want to replace the seating once every 7 years.

Critically, I don't think that this concept does enough to address these barriers to sustainability, it simply cannot offer enough to those with the power to change the monetary incentives.
This is a direct result of my focus on hands-on users, as optimising for their comfort means other areas of the full seating system suffer.

Longevity + Innovation

Physical Longevity

The significant innovations of this concept are the 'reshuffling' of structural components and the use of down/recycled wool throughout the design. Although these do not directly tackle issues with seat longevity, the simplified forms of the new frame and minimisation+combination of plastic components increase replaceability and promote replacement of components within use life.

Perceived Longevity

Similarly, the solutions presented in this concept do not directly tackle problems with visual modernity (**SPEC 2.2**). The design may appear modern and clean when new, but after wear and tear it will likely fall in line with current models.

Despite the improvements this concept makes to the replaceability and customisation, my concept doesn't make serious steps to ensure longevity beyond industry standards.

Appendix



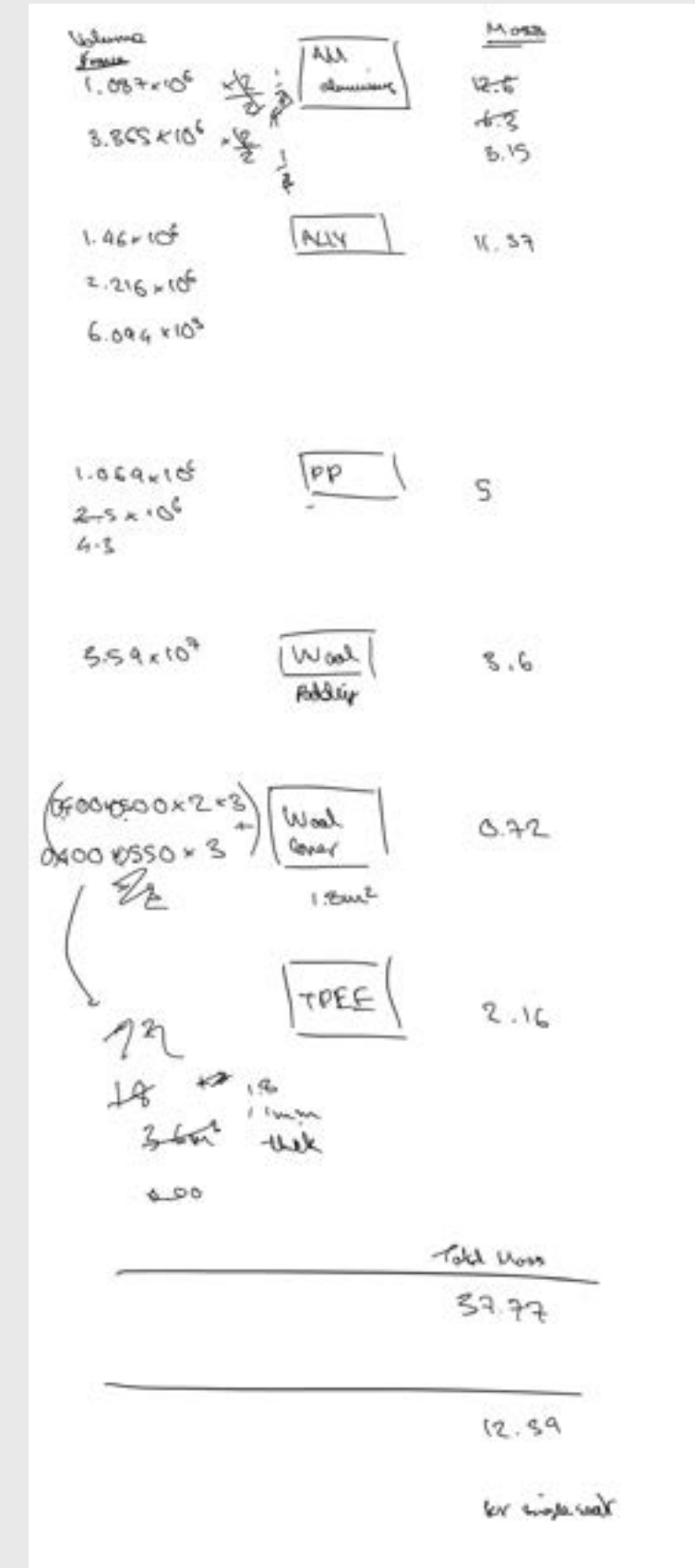
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Eco Audit Hand Calcs



First Final Concept Sketches

